

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_100018 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

partendo dal modello iniziale voglio modificare il materiale della struttura in modo che sia fatta di marmo

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Structural integrity compromised by material change

Subsystem: structure | Req: L1-MEC-002 | Type: constraint_violation

Marble is significantly less dense and structurally weaker than the original material, leading to potential failure under launch loads.

[MAJOR] ISSUE-002 — Shock test required due to material change

Subsystem: mechanical | Req: L1-MEC-003 | Type: ncr_required

The new material may not withstand the shock loads specified in the launcher SRS ICD, necessitating a retest.

[MINOR] ISSUE-003 — Thermal performance degradation due to material change

Subsystem: thermal | Req: L1-THM-001 | Type: reanalysis_required

Marble has different thermal properties, potentially increasing the ΔT across orbital cycles and violating the temperature stability requirement.

[MINOR] ISSUE-004 — Increased power consumption due to material change

Subsystem: power | Req: L1-PWR-002 | Type: reanalysis_required

The thermal properties of marble may require additional heaters or cooling systems, increasing the average power consumption.

MISSING ECSS TESTS — NOW REQUIRED

No additional tests required.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

No requirements directly invalidated.